

Title: Consistency Distillation as Augmented-Data ELBO at the Boundary

Hook: *The CD boundary instability isn't a numerical issue — it's a missing data augmentation. Consistency distillation IS the ELBO of an augmented model.*

Abstract

(missing)

Motivation

Problem framing. *(missing)*

Why now. *(missing)*

What changes when the gap closes. *(missing)*

Why prior work stopped: *(none)*

Core claim

Core claim. *(missing)*

Sub-claims. *(none)*

Method flow

High-level pipeline. *(missing)*

Steps. *(none)*

Preservation argument. *(missing)*

Theorem / algorithm / system design

(missing)

Theoretical leg

Theorem (Augmented-ELBO Equivalence): $L_{CD}(x_t, t) = ELBO_{\{A_\theta\}}(x_t, t) + \epsilon(t)$, where $\epsilon(t) \rightarrow 0$ as $\Delta t \rightarrow 0$ and $\epsilon(t) \leq C \cdot \|\nabla^2_{s_\theta}(x_t, t)\|_{op} \cdot \Delta t^2$ at finite NFE. Proof: Tweedie's formula applied to consistency-loss term with teacher-score-conditioned augmentation. Empirical falsification: tightness of $\epsilon(t)$ bound at finite NFE measured at multiple t-points.

Engineering leg

~8 lines of PyTorch replacing CD loss term with augmented-ELBO. Parameterize A_θ from teacher (existing teacher-score evaluation in CD training, no overhead), sample augmented data, use ELBO gradient. Rest of training pipeline byte-identical.

Falsification prediction

What we run. Train 4-NFE consistency-distilled student on CIFAR-10 with EDM teacher (50 NFE), comparing three setups: (a) vanilla CD, (b) SCott stochastic CD, (c) augmented-ELBO loss. Measure FID@4NFE on dev set and verify the equivalence theorem empirically by computing L_{CD} vs $L_{augmented_ELBO}$ on held-out batches at $t \in \{0.005, 0.05, 0.5\}$.

Success. FID@4NFE for augmented-ELBO student within 0.3 FID of EDM teacher@50NFE on CIFAR-10 dev; equivalence theorem's $\epsilon(t)$ bound holds within numerical precision ($\|L_{CD} - L_{augmented_ELBO}\| < \text{theoretical } \epsilon$ at all 3 t-points) for the augmented-ELBO student.

Failure (a) —numerical. FID@4NFE for augmented-ELBO student does not improve over vanilla CD or SCott (gap unchanged or worse).

Failure (b) —mechanistic. FID improves but the equivalence theorem’s empirical ϵ at boundary t ($t=0.005$) exceeds the theoretical bound — indicating the augmented-ELBO interpretation is wrong (the reformulation isn’t actually the loss being optimized; the FID gain came from regularization side-effect).

Compute budget. 0.6 GPU-day for the minimal experiment (CIFAR-10, single seed, 3 setups $\times$ 4 t -points probed). User has $4 \times A100 \times 3\text{wk} \approx 84$ A100-day; budget is 2 orders of magnitude under available — feasible.

Decision rule. if failure_a or failure_b $\rightarrow$ kill

Mechanism probe

Compute $L_{\text{CD}}(x_t, t)$ and $L_{\text{augmented_ELBO}}(x_t, t)$ on the same held-out batches at $t \in \{0.005, 0.05, 0.5\}$ — the equivalence theorem says they should agree within $\epsilon(t)$. Discriminating outcome: empirical $|L_{\text{CD}} - L_{\text{augmented_ELBO}}| \leq \epsilon_{\text{theoretical}}$ at each t for the augmented-ELBO student; for vanilla CD student the gap is unbounded at $t=0.005$. This distinguishes ‘reformulation IS the loss’ (success) from ‘reformulation is just a regularizer’ (failure_b).

Feasibility validation

Compute. (missing) —(missing)

Data. (missing) —(missing)

Theoretical. (missing) —(missing)

Engineering. (missing) —(missing)

Falsification. (missing) —(missing)

Overall. (missing)

Differentiation from recent literature

- **dblp:conf/aaai/LiuXYDCTFLZ25:** SCott introduces stochasticity into CD as a heuristic relaxation; we derive the relaxation as a specific A_θ that satisfies the ELBO equivalence — SCott’s stochasticity recovered as a special case of A_θ ’s noise schedule, not introduced as independent design choice.
- **dblp:conf/nips/0001LYLWLD0W24:** Motion CM disentangles motion vs appearance trajectories empirically; our derivation shows disentanglement is sufficient (lower Lipschitz at boundary satisfies our smoothness condition on A_θ) and need not be domain-specific to video.
- **arxiv:2603.08709v1:** Scale Space Diffusion connects diffusion timesteps to scale-space conceptually; we produce a closed-form algebraic identity with executable training change, not just conceptual reframing.
- **arxiv:2604.10857v1:** Query Lower Bounds gives information-theoretic minimum on score evaluations; our equivalence theorem characterizes when the consistency-loss objective at finite NFE is well-specified — lower bound says ‘few queries achievable in principle’, our theorem says ‘how to specify the loss to actually run there’.

Reviewer concerns and responses

(none)